



Chapter 5

Regression Analysis

"Regression analysis is the hydrogen bomb of the statistics arsenal."

— Charles Wheelan, *Naked Statistics: Stripping the Dread from the Data*

Regression analysis is a supervised learning method for predicting continuous variables. The difference between classification and regression analysis is that regression methods are used to predict qualitative variables or continuous numbers unlike categorical variables or labels. It is used to predict linear or non-linear relationships among variables of the given dataset. This chapter deals with an introduction of regression and its various types.

Learning Objectives

- Understand the basics of regression analysis
- Introduce concepts of correlation and causation
- Learn about linear regression and its validation techniques
- Discuss about multiple linear regression
- Introduce logistic regression
- Study about the concept of regularization
- Study popular regression methods like Ridge, Lasso, and Elastic Net

5.1 INTRODUCTION TO REGRESSION

Regression analysis is the premier method of supervised learning. This is one of the most popular and oldest supervised learning technique. Given a training dataset D containing N training points (x_i, y_i) , where $i = 1 \dots N$, regression analysis is used to model the relationship between one or more independent variables x_i and a dependent variable y_i . The relationship between the dependent and independent variables can be represented as a function as follows:

$$y = f(x) \quad (5.1)$$

Here, the feature variable x is also known as an explanatory variable, exploratory variable, a predictor variable, an independent variable, a covariate, or a domain point. y is a dependent variable. Dependent variables are also called as labels, target variables, or response variables.

Regression analysis determines the change in response variables when one exploration variable is varied while keeping all other parameters constant. This is used to determine the relationship each of the exploratory variables exhibits. Thus, regression analysis is used for prediction and forecasting.

Regression is used to predict continuous variables or quantitative variables such as price and revenue. Thus, the primary concern of regression analysis is to find answer to questions such as:

1. What is the relationship between the variables?
2. What is the strength of the relationships?
3. What is the nature of the relationship such as linear or non-linear?
4. What is the relevance of the attributes?
5. What is the contribution of each attribute?

There are many applications of regression analysis. Some of the applications of regressions include predicting:

1. Sales of a goods or services
2. Value of bonds in portfolio management
3. Premium on insurance companies
4. Yield of crops in agriculture
5. Prices of real estate

5.2 INTRODUCTION TO LINEARITY, CORRELATION, AND CAUSATION

The quality of the regression analysis is determined by the factors such as correlation and causation.

Regression and Correlation

Correlation among two variables can be done effectively using a Scatter plot, which is a plot between explanatory variables and response variables. It is a 2D graph showing the relationship between two variables. The x -axis of the scatter plot is independent, or input or predictor variables and y -axis of the scatter plot is output or dependent or predicted variables. The scatter plot is useful in exploring data. Some of the scatter plots are shown in Figure 5.1. The Pearson correlation coefficient is the most common test for determining correlation if there is an association between two variables. The correlation coefficient is denoted by r . Correlation is discussed in Chapter 2 of this book. The positive, negative, and random correlations are given in Figure 5.1. In positive correlation, one variable change is associated with the change in another variable. In negative correlation, the relationship between the variables is reciprocal while in random correlation, no relationship exists between variables.

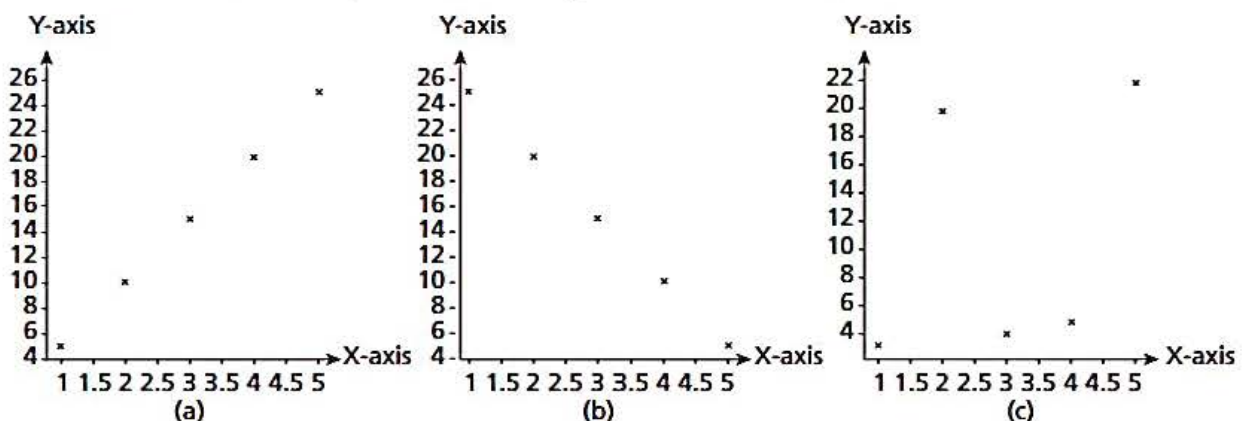


Figure 5.1: Examples of (a) Positive Correlation (b) Negative Correlation (c) Random Points with No Correlation

While correlation is about relationships among variables, say x and y , regression is about predicting one variable given another variable.

Regression and Causation

Causation is about causal relationship among variables, say x and y . Causation means knowing whether x causes y to happen or vice versa. x causes y is often denoted as x implies y . Correlation and Regression relationships are not same as causation relationship. For example, the correlation between economical background and marks scored does not imply that economic background causes high marks. Similarly, the relationship between higher sales of cool drinks due to a rise in temperature is not a causal relation. Even though high temperature is the cause of cool drinks sales, it depends on other factors too.

Linearity and Non-linearity Relationships

The linearity relationship between the variables means the relationship between the dependent and independent variables can be visualized as a straight line. The line of the form, $y = ax + b$ can be fitted to the data points that indicate the relationship between x and y . By linearity, it is meant that as one variable increases, the corresponding variable also increases in a linear manner. A linear relationship is shown in Figure 5.2 (a). A non-linear relationship exists in functions such as exponential function and power function and it is shown in Figures 5.2 (b) and 5.2 (c). Here, x -axis is given by x data and y -axis is given by y data.

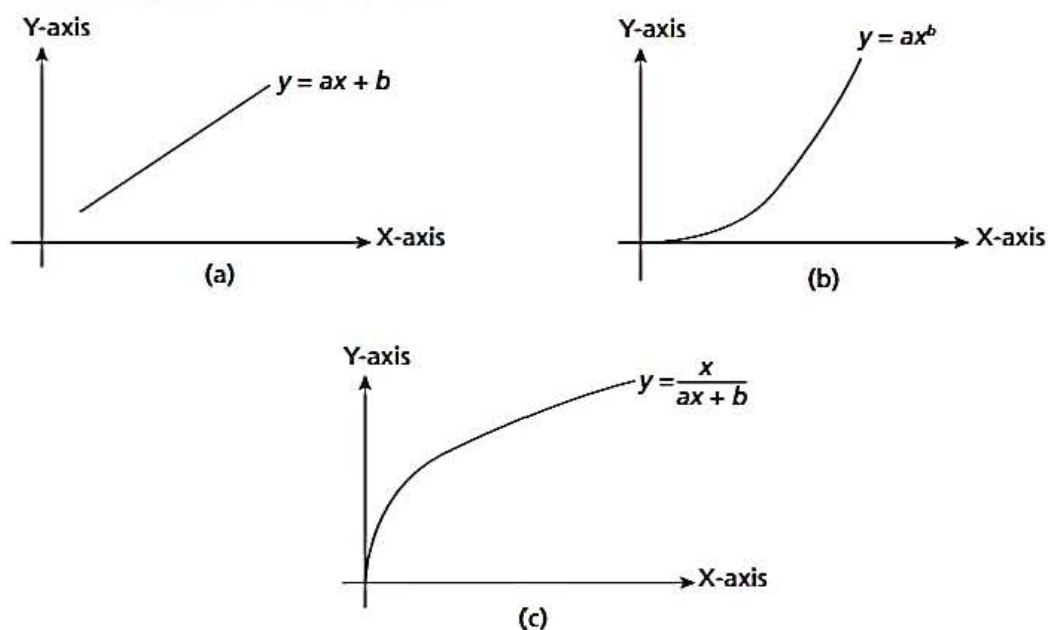


Figure 5.2: (a) Example of Linear Relationship of the Form $y = ax + b$ (b) Example of a Non-linear Relationship of the Form $y = ax^b$ (c) Examples of a Non-linear Relationship $y = \frac{x}{ax + b}$

The functions like exponential function ($y = ax^b$) and power function ($y = \frac{x}{ax + b}$) are non-linear relationships between the dependent and independent variables that cannot be fitted in a line. This is shown in Figures 5.2 (b) and (c).

Types of Regression Methods

The classification of regression methods is shown in Figure 5.3.

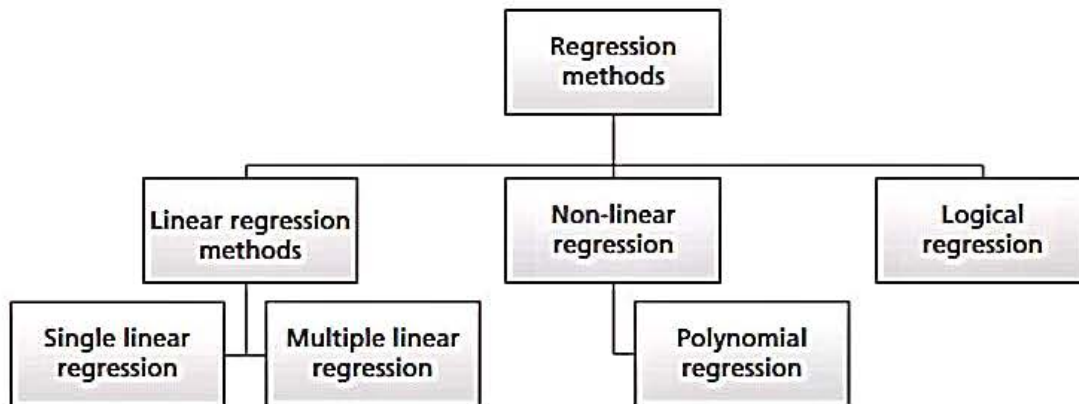


Figure 5.3: Types of Regression Methods

Linear Regression It is a type of regression where a line is fitted upon given data for finding the linear relationship between one independent variable and one dependent variable to describe relationships.

Multiple Regression It is a type of regression where a line is fitted for finding the linear relationship between two or more independent variables and one dependent variable to describe relationships among variables.

Polynomial Regression It is a type of non-linear regression method of describing relationships among variables where N^{th} degree polynomial is used to model the relationship between one independent variable and one dependent variable. Polynomial multiple regression is used to model two or more independent variables and one dependant variable.

Logistic Regression It is used for predicting categorical variables that involve one or more independent variables and one dependent variable. This is also known as a binary classifier.

Lasso and Ridge Regression Methods These are special variants of regression method where regularization methods are used to limit the number and size of coefficients of the independent variables.

Limitations of Regression Method

1. **Outliers** – Outliers are abnormal data. It can bias the outcome of the regression model, as outliers push the regression line towards it.
2. **Number of cases** – The ratio of independent and dependent variables should be at least 20 : 1. For every explanatory variable, there should be at least 20 samples. Atleast five samples are required in extreme cases.
3. **Missing data** – Missing data in training data can make the model unfit for the sampled data.
4. **Multicollinearity** – If exploratory variables are highly correlated (0.9 and above), the regression is vulnerable to bias. Singularity leads to perfect correlation of 1. The remedy is to remove exploratory variables that exhibit correlation more than 1. If there is a tie, then the tolerance ($1 - R$ squared) is used to eliminate variables that have the greatest value.

5.3 INTRODUCTION TO LINEAR REGRESSION

In the simplest form, the linear regression model can be created by fitting a line among the scattered data points. The line is of the form given in Eq. (5.2).

$$y = a_0 + a_1 \times x + e \quad (5.2)$$

Here, a_0 is the intercept which represents the bias and a_1 represents the slope of the line. These are called regression coefficients. e is the error in prediction.

The assumptions of linear regression are listed as follows:

1. The observations (y) are random and are mutually independent.
2. The difference between the predicted and true values is called an error. The error is also mutually independent with the same distributions such as normal distribution with zero mean and constant variables.
3. The distribution of the error term is independent of the joint distribution of explanatory variables.
4. The unknown parameters of the regression models are constants.

The idea of linear regression is based on Ordinary Least Square (OLS) approach. This method is also known as ordinary least squares method. In this method, the data points are modelled using a straight line. Any arbitrarily drawn line is not an optimal line. In Figure 5.4, three data points and their errors (e_1, e_2, e_3) are shown. The vertical distance between each point and the line (predicted by the approximate line equation $y = a_0 + a_1x$) is called an error. These individual errors are added to compute the total error of the predicted line. This is called sum of residuals. The squares of the individual errors can also be computed and added to give a sum of squared error. The line with the lowest sum of squared error is called line of best fit.

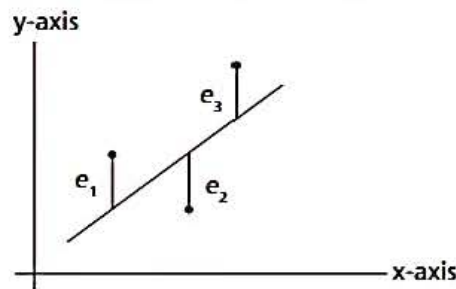


Figure 5.4: Data Points and their Errors

In another words, OLS is an optimization technique where the difference between the data points and the line is optimized.

Mathematically, based on Eq. (5.2), the line equations for points ($x_1, x_2, \dots, x_n$) are:

$$\begin{aligned} y_1 &= (a_0 + a_1x_1) + e_1 \\ y_2 &= (a_0 + a_1x_2) + e_2 \\ &\vdots \\ &\vdots \\ &\vdots \\ y_n &= (a_0 + a_1x_n) + e_n \end{aligned} \quad (5.3)$$

$$\text{In general, the error is given as: } e_i = y_i - (a_0 + a_1x_i) \quad (5.4)$$

This can be extended into the set of equations as shown in Eq. (5.3).

Here, the terms $(e_1, e_2, \dots, e_n)$ are error associated with the data points and denote the difference between the true value of the observation and the point on the line. This is also called as residuals. The residuals can be positive, negative or zero.

A regression line is the line of best fit for which the sum of the squares of residuals is minimum. The minimization can be done as minimization of individual errors by finding the parameters a_0 and a_1 such that:

$$E = \sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - (a_0 + a_1 x_i)) \quad (5.5)$$

Or as the minimization of sum of absolute values of the individual errors:

$$E = \sum_{i=1}^n |e_i| = \sum_{i=1}^n |(y_i - (a_0 + a_1 x_i))| \quad (5.6)$$

Or as the minimization of the sum of the squares of the individual errors:

$$E = \sum_{i=1}^n (e_i)^2 = \sum_{i=1}^n (y_i - (a_0 + a_1 x_i))^2 \quad (5.7)$$

Sum of the squares of the individual errors, often preferred as individual errors (positive and negative errors), do not get cancelled out and are always positive, and sum of squares results in a large increase even for a small change in the error. Therefore, this is preferred for linear regression.

Therefore, linear regression is modelled as a minimization function as follows:

$$\begin{aligned} J(a_1, a_0) &= \sum_{i=1}^n [y_i - f(x_i)]^2 \\ &= \sum_{i=1}^n [y_i - (a_0 + a_1 x_i)]^2 \end{aligned} \quad (5.8)$$

Here, $J(a_0, a_1)$ is the criterion function of parameters a_0 and a_1 . This needs to be minimized. This is done by differentiating and substituting to zero. This yields the coefficient values of a_0 and a_1 . The values of estimates of a_0 and a_1 are given as follows:

$$a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{(\overline{x^2}) - (\bar{x})^2} \quad (5.9)$$

And the value of a_0 is given as follows:

$$a_0 = (\bar{y}) - a_1 \times \bar{x} \quad (5.10)$$

Let us consider a simple problem to illustrate the usage of the above concept.

Example 5.1: Let us consider an example where the five weeks' sales data (in Thousands) is given as shown below in Table 5.1. Apply linear regression technique to predict the 7th and 9th month sales.

Table 5.1: Sample Data

x_i (Week)	y_i (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Solution: Here, there are 5 items, i.e., $i = 1, 2, 3, 4, 5$. The computation table is shown below (Table 5.2). Here, there are five samples, so i ranges from 1 to 5.

Table 5.2: Computation Table

x_i	y_i	$(x_i)^2$	$x_i \times y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15	Sum = 12.6	Sum = 55	Sum = 44.4
Average of (x_i)	Average of (y_i)	Average of (x_i^2)	Average of $(x_i \times y_i)$
$= \bar{x} = \frac{15}{5}$	$= \bar{y} = \frac{12.6}{5}$	$= \overline{x_i^2} = \frac{55}{5}$	$= \overline{xy} = \frac{44.4}{5}$
$= 3$	$= 2.52$	$= 11$	$= 8.88$

Let us compute the slope and intercept now using Eq. (5.9) as:

$$a_1 = \frac{8.88 - 3(2.52)}{11 - 3^2} = 0.66$$

$$a_0 = 2.52 - 0.66 \times 3 = 0.54$$

The fitted line is shown in Figure 5.5.

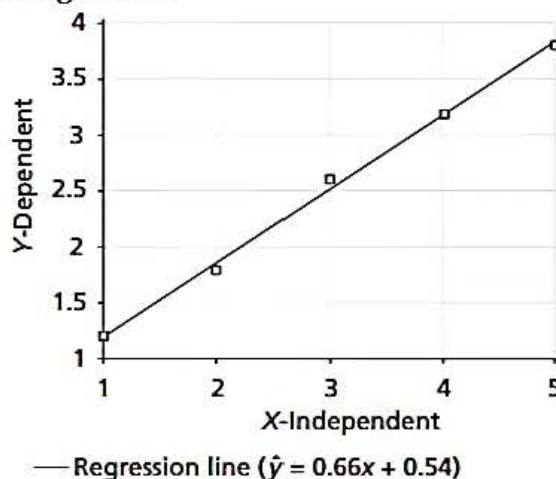


Figure 5.5: Linear Regression Model Constructed

Let us model the relationship as $y = a_0 + a_1 \times x$. Therefore, the fitted line for the above data is:
 $y = 0.54 + 0.66 \times x$.

The predicted 7th week sale would be (when $x = 7$), $y = 0.54 + 0.66 \times 7 = 5.16$ and the 12th month, $y = 0.54 + 0.66 \times 12 = 8.46$. All sales are in thousands.

Scan for 'Additional Examples'



Linear Regression in Matrix Form

Matrix notations can be used for representing the values of independent and dependent variables. This is illustrated through Example 5.2.

The Eq. (5.3) can be written in the form of matrix as follows:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} \quad (5.11)$$

This can be written as:

$Y = Xa + e$, where X is an $n \times 2$ matrix, Y is an $n \times 1$ vector, a is a 2×1 column vector and e is an $n \times 1$ column vector.

Example 5.2: Find linear regression of the data of week and product sales (in Thousands) given in Table 5.3. Use linear regression in matrix form.

Table 5.3: Sample Data for Regression

x_i (Week)	y_i (Product Sales in Thousands)
1	1
2	3
3	4
4	8

Solution: Here, the dependent variable X is be given as:

$$x^T = [1 \ 2 \ 3 \ 4]$$

And the independent variable is given as follows:

$$y^T = [1 \ 3 \ 4 \ 8]$$

The data can be given in matrix form as follows:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}. \text{ The first column can be used for setting bias.}$$

$$\text{and } Y = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 8 \end{pmatrix}$$

The regression is given as:

$$a = ((X^T X)^{-1} X^T) Y$$

The computation order of this equation is shown step by step as:

1. Computation of $(X^T X) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \times \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}$
2. Computation of matrix inverse of $(X^T X)^{-1} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}^{-1} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix}$
3. Computation of $((X^T X)^{-1} X^T) = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0.5 & 0 & -0.5 \\ -0.3 & -0.1 & 0.1 & 0.3 \end{pmatrix}$
4. Finally, $((X^T X)^{-1} X^T) Y = \begin{pmatrix} 1 & 0.5 & 0 & -0.5 \\ -0.3 & -0.1 & 0.1 & 0.3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} -1.5 \\ 2.2 \end{pmatrix} \begin{pmatrix} \text{Intercept} \\ \text{slope} \end{pmatrix}$

Thus, the substitution of values in Eq. (5.11) using the previous steps yields the fitted line as $2.2x - 1.5$.

5.4 VALIDATION OF REGRESSION METHODS

The regression model should be evaluated using some metrics for checking the correctness. The following metrics are used to validate the results of regression.

Standard Error

Residuals or error is the difference between the actual (y) and predicted value ($\hat{y}$).

If the residuals have normal distribution, then the mean is zero and hence it is desirable. This is a measure of variability in finding the coefficients. It is preferable that the error be less than the coefficient estimate. The standard deviation of residuals is called residual standard error. If it is zero, then it means that the model fits the data correctly.

Mean Absolute Error (MAE)

MAE is the mean of residuals. It is the difference between estimated or predicted target value and actual target incomes. It can be mathematically defined as follows:

$$MAE = \frac{1}{n} \sum_{i=0}^{n-1} |y_i - \hat{y}_i| \quad (5.12)$$

Here, $\hat{y}$ is the estimated or predicted target output and y is the actual target output, and n is the number of samples used for regression analysis.

Mean Squared Error (MSE)

It is the sum of square of residuals. This value is always positive and closer to 0. This is given mathematically as:

$$\frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2 \quad (5.13)$$

Root Mean Square Error (RMSE)

The square root of the MSE is called RMSE. This is given as:

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2} \quad (5.14)$$

Relative MSE

Relative MSE is the ratio of the prediction ability of the $\hat{y}$ to the average of the trivial population. The value of zero indicates that the model is perfect and its value ranges between 0 and 1. If the value is more than 1, then the created model is not a good one. This is given as follows:

$$\text{RelMSE} = \frac{\sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2}{\sum_{i=0}^{n-1} (y_i - \bar{y}_i)^2} \quad (5.15)$$

Coefficient of Variation

Coefficient of variation is unit less and is given as:

$$\text{CV} = \frac{\text{RMSE}}{\bar{y}} \quad (5.16)$$

Example 5.3: Consider the following training set Table 5.4 for predicting the sales of the items.

Table 5.4: Training Item Table

Items x_i	Actual Sales (In Thousands) y_i
I_1	80
I_2	90
I_3	100
I_4	110
I_5	120

Consider two fresh items I_6 and I_7 , whose actual values are 80 and 75, respectively. A regression model predicts the values of the items I_6 and I_7 as 75 and 85, respectively. Find MAE, MSE, RMSE, RelMSE and CV.

Solution: The test items' actual and prediction is given in Table 5.5 as:

Table 5.5: Test Item Table

Test Items	Actual Value y_i	Predicted Value $\bar{y}_i$
I_6	80	75
I_7	75	85

Mean Absolute Error (MAE) using Eq. (5.12) is given as:

$$\text{MAE} = \frac{1}{2} \times |80 - 75| + |75 - 85| = \frac{15}{2} = 7.5$$

Mean Squared Error (MSE) using Eq. (5.13) is given as:

$$\text{MSE} = \frac{1}{2} \times |80 - 75|^2 + |75 - 85|^2 = \frac{125}{2} = 62.5$$

Root Mean Square error using Eq. (5.14) is given as:

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{62.5} = 7.91$$

For finding RelMSE and CV, the training table should be used to find the average of y .

The average of y is $\frac{80 + 90 + 100 + 110 + 120}{5} = \frac{500}{5} = 100$.

RelMSE using Eq. (5.15) can be computed as:

$$\text{RelMSE} = \frac{(80 - 75)^2 + (75 - 85)^2}{(80 - 100)^2 + (75 - 100)^2} = \frac{125}{1025} = 0.1219$$

CV can be computed using Eq. (5.16) as $\frac{\sqrt{62.5}}{100} = 0.08$.

Coefficient of Determination

To understand the coefficient of determination, one needs to understand the total variation of coefficients in regression analysis. The sum of the squares of the differences between the y -value of the data pair and the average of y is called total variation. Thus, the following variations can be defined.

The explained variation is given as:

$$= \sum (\hat{y}_i - \bar{y})^2 \quad (5.17)$$

The unexplained variation is given as:

$$= \sum (y_i - \hat{y}_i)^2 \quad (5.18)$$

Thus, the total variation is equal to the explained variation and the unexplained variation.

The coefficient of determination r^2 is the ratio of the explained and total variations.

$$r^2 = \frac{\text{Explained variation}}{\text{Total variation}} \quad (5.19)$$

It is a measure of how many future samples are likely to be predicted by the regression model. Its value ranges from 1 to $-\infty$, where 1 is the most optimum value. It also signifies the proportion of variance. Here, r is the correlation coefficient. If $r = 0.95$, then r^2 is given as $0.95 \times 0.95 = 0.9025$. This means that 90% of the model can be explained by the relationship between x and y . The rest 10% is unexplained and that may be due to various reasons such as noise, chance, or error.

Standard Error Estimate

Standard error estimate is another useful measure of regression. It is the standard deviation of the observed values to the predicted values. This is given as:

$$s_e = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}} \quad (5.20)$$

Here, as usual, y_i is the observed value and $\hat{y}_i$ is the predicted value. Here, n is the number of samples.

Example 5.4: Let us consider the data given in the Table 5.3 with actual and predicted values. Find standard error estimate.

Solution: The observed value or the predicted value is given below in Table 5.6.

Table 5.6: Sample Data

x_i	y_i	Predicted Value	$(y - \hat{y})^2$
1	1.5	1.46	$(1.5 - 1.46)^2 = 0.0016$
2	2.9	2.02	$(2.9 - 2.02)^2 = 0.7744$
3	2.7	2.58	$(2.7 - 2.58)^2 = 0.0144$
4	3.1	3.14	$(3.1 - 3.14)^2 = 0.0016$

The sum of $(y - \hat{y})^2$ for all $i = 1, 2, 3$ and 4 (i.e., number of samples $n = 4$) is 0.792. The standard deviation error estimate as given in Eq. (5.20) is:

$$\sqrt{\frac{0.792}{4 - 2}} = \sqrt{0.396} = 0.629$$

5.5 MULTIPLE LINEAR REGRESSION

Multiple regression model involves multiple predictors or independent variables and one dependent variable. This is an extension of the linear regression problem. The basic assumptions of multiple linear regression are that the independent variables are not highly correlated and hence multicollinearity problem does not exist. Also, it is assumed that the residuals are normally distributed.

For example, the multiple regression of two variables x_1 and x_2 is given as follows:

$$\begin{aligned} y &= f(x_1, x_2) \\ &= a_0 + a_1x_1 + a_2x_2 \end{aligned} \quad (5.21)$$

In general, this is given for ' n ' independent variables as:

$$\begin{aligned} y &= f(x_1, x_2, x_3, \dots, x_n) \\ &= a_0 + a_1x_1 + a_2x_2 + \dots + a_nx_n + \varepsilon \end{aligned} \quad (5.22)$$

Here, $(x_1, x_2, \dots, x_n)$ are predictor variables, y is the dependent variable, $(a_0, a_1, \dots, a_n)$ are the coefficients of the regression equation and ε is the error term. This is illustrated through Example 5.5.

Example 5.5: Apply multiple regression for the values given in Table 5.7 where weekly sales along with sales for products x_1 and x_2 are provided. Use matrix approach for finding multiple regression.

Table 5.7: Sample Data

x_1 (Product One Sales)	x_2 (Product Two Sales)	y Output Weekly Sales (in Thousands)
1	4	1
2	5	6
3	8	8
4	2	12

Solution: Here, the matrices for Y and X are given as follows:

$$X = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} \text{ and } Y = \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}.$$

The coefficient of the multiple regression equation is given as $a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$.

The regression coefficient for multiple regression is calculated the same way as linear regression:

$$\hat{a} = ((X^T X)^{-1} X^T) Y \quad (5.23)$$

Using Eq. (5.23), and substituting the values (Similar to Problem 5.2), one gets $\hat{a}$ as:

$$\begin{aligned} &= \left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} \right)^{-1} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix} \\ &= \begin{pmatrix} -1.69 \\ 3.48 \\ -0.05 \end{pmatrix} \end{aligned}$$

Here, the coefficients are $a_0 = -1.69$, $a_1 = 3.48$ and $a_2 = -0.05$. Hence, the constructed model is:
 $y = -1.69 + 3.48x_1 - 0.05x_2$

5.6 POLYNOMIAL REGRESSION

If the relationship between the independent and dependent variables is not linear, then linear regression cannot be used as it will result in large errors. The problem of non-linear regression can be solved by two methods:

1. Transformation of non-linear data to linear data, so that the linear regression can handle the data
2. Using polynomial regression

Transformations

The first method is called transformation. The trick is to convert non-linear data to linear data that can be handled using the linear regression method. Let us consider an exponential function $y = ae^{bx}$. The transformation can be done by applying log function to both sides to get:

$$\ln y = bx + \ln a \quad (5.24)$$

Similarly, power function of the form $(y = ax^b)$ can be transformed by applying log function on both sides as follows:

$$\log_{10} y = b \log_{10} x + \log_{10} a \quad (5.25)$$

Once the transformation is carried out, linear regression can be performed and after the results are obtained, the inverse functions can be applied to get the desired result.

Polynomial Regression

It can handle non-linear relationships among variables by using n^{th} degree of a polynomial. Instead of applying transforms, polynomial regression can be directly used to deal with different levels of curvilinearity.

Polynomial regression provides a non-linear curve such as quadratic and cubic. For example, the second-degree transformation (called quadratic transformation) is given as: $y = a_0 + a_1x + a_2x^2$ and the third-degree polynomial is called cubic transformation given as: $y = a_0 + a_1x + a_2x^2 + a_3x^3$. Generally, polynomials of maximum degree 4 are used, as higher order polynomials take some strange shapes and make the curve more flexible. It leads to a situation of overfitting and hence is avoided.

Let us consider a polynomial of 2nd degree. Given points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the objective is to fit a polynomial of degree 2. The polynomial of degree 2 is given as:

$$y = a_0 + a_1x + a_2x^2 \quad (5.26)$$

Such that the error $E = \sum_{i=1}^n [y_i - (a_0 + a_1x_i + a_2x_i^2)]^2$ is minimized. The coefficients a_0, a_1, a_2 of Eq. (5.26) can be obtained by taking partial derivatives with respect to each of the coefficients as $\frac{\partial E}{\partial a_0}, \frac{\partial E}{\partial a_1}, \frac{\partial E}{\partial a_2}$ and substituting it with zero. This results in 2 + 1 equations given as follows:

$$\begin{aligned} na_0 + \left(\sum_{i=1}^n x_i \right) a_1 + \left(\sum_{i=1}^n x_i^2 \right) a_2 &= \sum_{i=1}^n y_i \\ \left(\sum_{i=1}^n x_i \right) a_0 + \left(\sum_{i=1}^n x_i^2 \right) a_1 + \left(\sum_{i=1}^n x_i^3 \right) a_2 &= \sum_{i=1}^n x_i y_i \\ \left(\sum_{i=1}^n x_i^2 \right) a_0 + \left(\sum_{i=1}^n x_i^3 \right) a_1 + \left(\sum_{i=1}^n x_i^4 \right) a_2 &= \sum_{i=1}^n x_i^2 y_i \end{aligned} \quad (5.27)$$

The best line is the line that minimizes the error between line and data points. Arranging the coefficients of the above equation in the matrix form results in:

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i, y_i) \\ \sum (x_i^2, y_i) \end{bmatrix} \quad (5.28)$$

This is of the form $Xa = B$. One can solve this equation for a as:

$$a = X^{-1}B \quad (5.29)$$

Example 5.6: Consider the data provided in Table 5.8 and fit it using the second-order polynomial.

Table 5.8: Sample Data

x	y
1	1
2	4
3	9
4	15

Solution: For applying polynomial regression, computation is done as shown in Table 5.9.

Here, the order is 2 and the sample i ranges from 1 to 4.

Table 5.9: Computation Table

x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y_i$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\Sigma x_i = 10$	$\Sigma y_i = 29$	$\Sigma x_i y_i = 96$	$\Sigma x_i^2 = 30$	$\Sigma x_i^2 y_i = 338$	$\Sigma x_i^3 = 100$	$\Sigma x_i^4 = 354$

It can be noted that, $N = 4$, $\Sigma y_i = 29$, $\Sigma x_i y_i = 96$, $\Sigma x_i^2 y_i = 338$. When the order is 2, the matrix using Eq. (5.28) is given as follows:

$$\begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix}$$

Therefore, using Eq. (5.29), one can get coefficients as:

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{bmatrix} -0.75 \\ 0.95 \\ 0.75 \end{bmatrix}.$$

This leads to the regression equation using Eq. (5.26) as:

$$y = -0.75 + 0.95x + 0.75x^2$$

5.7 LOGISTIC REGRESSION

Linear regression predicts the numerical response but is not suitable for predicting the categorical variables. When categorical variables are involved, it is called classification problem. Logistic regression is suitable for binary classification problem. Here, the output is often a categorical variable. For example, the following scenarios are instances of predicting categorical variables.

1. Is the mail spam or not spam? The answer is yes or no. Thus, categorical dependant variable is a binary response of yes or no.
2. If the student should be admitted or not is based on entrance examination marks. Here, categorical variable response is admitted or not.
3. The student being pass or fail is based on marks secured.

Thus, logistic regression is used as a binary classifier and works by predicting the probability of the categorical variable. In general, it takes one or more features x and predicts the response y . If the probability is predicted via linear regression, it is given as:

$$p(x) = a_0 + a_1x$$

Hence, logistic regression tries to model the probability of the particular response variable. In email classification problem, say normal email or spam, if the probability of the response variable is 0.7, then there is a 70% possibility of a normal mail.

Linear regression generated value is in the range $-\infty$ to $+\infty$, whereas the probability of the response variable ranges between 0 and 1. Hence, there must be a mapping function to map the value $-\infty$ to $+\infty$ to 0–1. The core of the mapping function in logistic regression method is sigmoidal function. A sigmoidal function is a 'S' shaped function that yields values between 0 and 1. This is known as logit function. This is mathematically represented as:

$$\text{logit}(x) = \frac{1}{1 + e^{-x}} \quad (5.30)$$

Here, x is the independent variable and e is the Euler number. The purpose of the logit function is to map any real number to zero or 1.

Logistic regression can be viewed as an extension of linear regression, but the only difference is that the output of linear regression can be an extremely high number. This needs to be mapped into the range 0–1, as probability can have values only in the range 0–1. This problem is solved using log odd or logit functions. What is the difference between odds and probability? Odds and probability (or likelihood) are two sides of a coin and represent uncertainty. The odds are defined as the ratio of the probability of an event and probability of an event that is not happening. This is given as:

$$\text{odd} = \frac{\text{probability of an event}}{\text{probability of an non-event}} = \frac{p}{1-p} \quad (5.31)$$

Log-odds can be taken for the odds, resulting in:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = a_0 + a_1x \quad (5.32)$$

Here, $\log(\cdot)$ is a logit function or log odds function. One can solve for $p(x)$ by taking the inverse of the above function as:

$$p(x) = \frac{\exp(a_0 + a_1x)}{1 + \exp(a_0 + a_1x)} \quad (5.33)$$

This is the same sigmoidal function. It always gives the value in the range 0–1. Dividing the numerator and denominator by the numerator, one gets:

$$p(x) = \frac{1}{1 + \exp(-a_0 - a_1x)} \quad (5.34)$$

One can rearrange this by taking the minus sign outside to get the following logistic function:

$$p(x) = \frac{1}{1 + \exp(-(a_0 + a_1x))} \quad (5.35)$$

Here, x is the explanatory or predictor variable, e is the Euler number, and a_0, a_1 are the regression coefficients. The coefficients a_0, a_1 can be learned and the predictor predicts $p(x)$ directly using the threshold function as:

$$y = \begin{cases} 1 & \text{if } p(x) \geq 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (5.36)$$

Example 5.7: Let us assume a binomial logistic regression problem where the classes are pass and fail. The student dataset has entrance mark based on the historic data of those who are selected or not selected. Based on the logistic regression, the values of the learnt parameters are $a_0 = 1$ and $a_1 = 8$. Assuming marks of $x = 60$, compute the resultant class.

Solution: The values of regression coefficients are $a_0 = 1$ and $a_1 = 8$, and given that $x = 60$.

Based on the regression coefficients, z can be computed as:

$$\begin{aligned} z &= a_0 + a_1 x \\ &= 1 + 8 \times 60 = 481 \end{aligned}$$

One can fit this in a sigmoidal function using Eq. (5.30) to get the probability as:

$$y = \frac{1}{1 + \exp(-481)} = \frac{1}{2.271} = 0.44$$

If we assume the threshold value as 0.5, then it is observed that $0.44 < 0.5$, therefore, the candidate with marks 60 is not selected.

To determine the relationship between dependant and independent variables, parameters need to be obtained. In logistic regression, the parameters are obtained through maximum likelihood function (MLE) using the training data. The aim is to learn the values of parameters of the logistic model (a 's) by minimizing the error in the probability predicted by the model.

There can be many different sets of coefficients available. The optimal value of the parameters is obtained by using the MLE function, which is a set of coefficients for which the probability of getting the observed data is maximum.

If π is the success of the outcome and $1 - \pi$ is the failure of the outcome, then the likelihood function is given as:

$$L(a : y) = \prod_{i=1}^n \left(\frac{\pi_i}{1 - \pi_i} \right)^{y_i} (1 - \pi_i) \quad (5.37)$$

To determine the value of the parameters, the log of the likelihood function is taken. Techniques like Newton method can be used to maximize the log-likelihood of the function.

Logistic regression is suitable for binary classification. The idea can be extended for multiple classes called multinomial logistic regression. Let us assume that there are three classes 1, 2 and 3. Then, the multinomial logistic regression creates three classification problems – class 1 and Not class 1, class 2 and Not class 2, and finally class 3 and Not class 3. Three problems are simultaneously used to find the maximum probability relative to others to get the appropriate class.

Logistic regression is a simple and efficient method for binary classification. The model can be easily interpreted too. The disadvantages of logistic regression are that multinomial logistic regression cannot handle many attributes and can handle only linear features. Also, if all the attributes have multicollinearity problem, the logistic method does not work effectively.

5.8 RIDGE, LASSO, AND ELASTIC NET REGRESSION

Recollect that Ordinary Least Square (OLS) fits a line for the data points that minimize the sum of squared error between the data points and the line of fit.

There are two issues that need to be considered before discussing regularization methods. One is bias and another is variance.

1. Bias – If the selected model does not fit the dataset well, then this error is called bias.
2. Variance – If the model works nicely for the trained data but is not representative for the entire universe of the possible data, it is called variance.

Consider the Figure 5.6 where the errors of the individual points are shown.

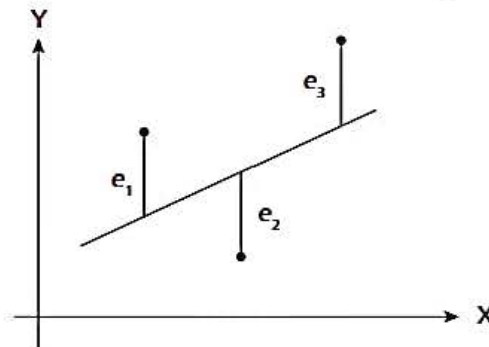


Figure 5.6: The Variance of the Model

The vertical line that measures the error between the data points and the line of fit indicates the variance. The sum of the individual error squares refers to the amount of variance that is not captured by the model. It should be minimal.

If the line fits the training points but lacks the ability to fit the testing phase points, it indicates generalization error. A line of good fit should have generalization capability and should have the least error for the training as well as testing phase. A line that minimizes the error for the training points and more error for the testing phase is called overfitting problem. Overfitting gives more variance.

The main idea to solve variance is to introduce a small bias to create a ridge line to reduce variance. Consider the following Figure 5.7. Here, the first two points are training data points and the next two points are test data points. If a line is fit through the training data, then it may not work properly for the test data points. In other words, the model lacks generality.

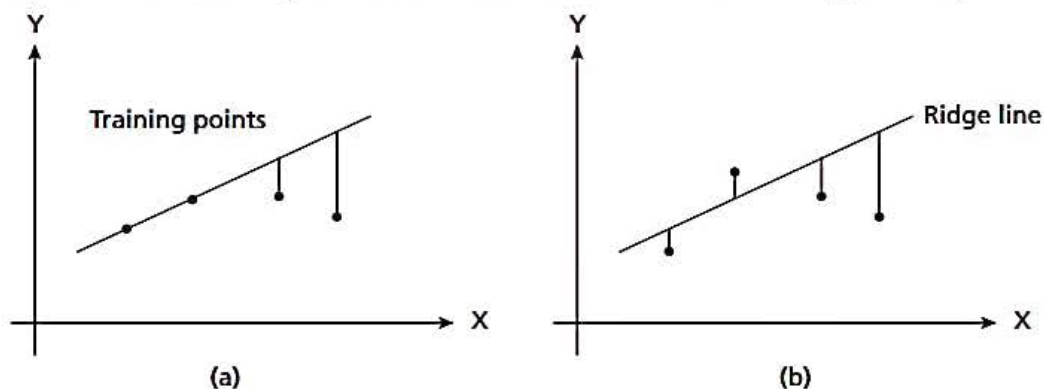


Figure 5.7: (a) OLS Fit Line (b) Ridge Line

The solution for this problem is to fit a ridge line by changing the slope of the OLS line. This introduces a bias in the model and because of the bias, even though this line does not fit the training data, the overall variance gets reduced. So, the aim of regularization is to reduce the test data error.

Normally, low variance algorithm has high bias and vice versa. If we have high variance and low bias, the problem is called underfitting. The reciprocal problem is called overfitting. So, there must be a balance between bias and variance. This can be achieved by removing the unnecessary attributes of the dataset. This is done by two ways:

1. Feature selection algorithms remove unnecessary attributes. The forward selection and backward selection algorithms and principle component analysis are discussed in Chapter 2.
2. Another way to solve the problem is to use regularization methods that add a penalty to penalize the regression coefficients to simplify the model.

Let us discuss about regularization methods now.

5.8.1 Ridge Regularization

Ridge regression is used to create a parsimonious model. It is useful when there are many attributes that exceed the number of observations with strong correlation among them. Ridge regression is a plain, linear regression model whose regression coefficients are not estimated by OLS but by a ridge estimator. Ridge estimator is a biased estimator that has lower variance than the OLS estimator. The mean square error of the ridge estimator (sum of the variance and square of its bias) is lower than the estimator of OLS method. The estimator for ridge regression is given as follows:

$$\text{Sum of squared residuals} + \lambda \times \text{slope}^2$$

$$\text{That is, } \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \times \text{slope}^2 \quad (5.38)$$

Here, $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ is the sum of the squared residuals and $\lambda \times \text{slope}^2$ is the penalty term added to change the slope of the line of fit to reduce variance. The purpose of the penalty is to penalize the large regression coefficients. If the value of the regression is more, then penalty is more. This is done by reducing the variance by adding a small bias. The change of slope is the main aspect of ridge regression. It reduces the slope of the line. If the slope of the line is small, then the change in prediction is barely noticeable. Thus, ridge line creates a small slope and is hence insensitive.

Ridge regression uses L_2 regularization. This limits the size of the coefficients using a penalty function. L_2 Penalty equals the square of the magnitude of all coefficients. The penalty is controlled by a parameter λ . When $\lambda = 0$, ridge regression is same as OLS. When λ is infinity, the slope becomes almost zero and hence shrinks all coefficients to 0. Thus, ridge regression avoids the problem of overfitting by shrinking the less important variables closer to zero. The choice of λ is very crucial. It is obtained by leave-one-out validation method. The details of this validation method are discussed in Chapter 3. The procedure for selecting λ is given as follows:

1. Choose a possible set of values of the penalty
2. Exclude the N^{th} observed data and compute the penalty

3. Compute the penalty out of samples that are excluded
4. Compute the mean square error and pick the penalty that minimizes the MSE

The matrix formulation of ridge regression is given as follows:

$$\hat{a} = (X'X + kI)^{-1} X'Y \quad (5.39)$$

The major problem is the selection of ' k ' in Eq. (5.39). Ridge regression is also useful for logistic regression where it is expressed as sum of likelihoods + $\lambda \times \text{slope}^2$. This idea can be extended for many independent variables' regression coefficients except y -intercept.

5.8.2 LASSO

LASSO regression is better than ridge regression. LASSO stands for 'Least Absolute Shrinkage and Selection Operator'. This uses L_1 regularization.

This adds a penalty factor that equals the square of the magnitude of all coefficients. This penalty results in a simple model given as:

$$\text{Sum of squared residuals} + \lambda \times |\text{slope}|$$

Thus, LASSO model is given as:

$$\hat{a} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=0}^k |a_i| \quad (5.40)$$

By controlling the factor λ , where the high values of λ force many coefficients to zero, LASSO performs both shrinkage and variable selection.

The difference between Ridge and LASSO regression methods is given in the following Table 5.10.

Table 5.10: Difference between Ridge and LASSO Regression

Ridge Regression	LASSO Regression
Penalty term is $\lambda \times \text{slope}^2$	Penalty term is $\lambda \times \text{slope} $
This method shrinks the regression coefficients of less important variables closer to zero.	This method shrinks the regression coefficients of less important variables to zero giving a compact model.
Not useful for feature selection	Good feature selector by removing all irrelevant variables

5.8.3 Elastic Net

Elastic Net is a hybrid method of combining both Ridge and LASSO regression methods. The Elastic Net is given as follows:

$$\text{Sum of squared residuals} + \underbrace{\lambda_1 (|v_1| + |v_2| + \dots + |v_k|)}_{\text{Lasso}} + \underbrace{\lambda_2 (v_1^2 + v_2^2 + \dots + v_k^2)}_{\text{Ridge}} \quad (5.41)$$

Here, $v_1, v_2, \dots, v_k$ are dependent variables of the regression method.

It uses separate penalty factors for Lasso and Ridge regression methods. When λ_1 and λ_2 are zero, then Elastic Net reduces to simple OLS method. When $\lambda_1 = 0$, Elastic Net serves as ridge regression and when $\lambda_2 = 0$, Elastic Net serves as LASSO regression. When both λ_1 and λ_2 are greater than zero, then it serves as hybrid technique.

Elastic Net groups the variables and shrinks the parameters associated with the correlated variables and makes it closer to zero or zero, thereby removing the irrelevant attributes. Elastic Net is good in situations where multicollinearity problem exists among the independent variables of the regression method.

Summary

1. Regression is a supervised learning method that can predict continuous variables.
2. Regression analysis is used to model the relationship between one or more independent variables and a dependent variable whereas in multiple regression problems, the output is a combination of predictor variables.
3. Regression is used for prediction and forecasting. This determines the change in response variables when one exploration variable is varied while keeping all other parameters constant. This is used to determine the relationship each of the exploratory variables exhibits.
4. Scatter plot is a plot of explanatory variable and response variable. It is a 2D graph showing the relationship between two variables. The quality of the regression analysis is determined by the factors of correlation and causation.
5. Linear regression model can be created by fitting a line among the scattered data points. The line is of the form: $y = a_0 + a_1x$. Here, a_0 is the intercept which represents the bias and a_1 represents the slope of the line.
6. Multiple regression model is an extension of linear regression model that involves multiple predictors. It is also used to find relationships among variables, important features and construct models.
7. Polynomial regression can handle non-linear relationships among variables by using degrees of polynomial to make non-regression line. If the relationship is not linear, then polynomial regression can be used at different levels of curvilinearity.
8. Logistic regression is suitable for binary classification problem. Here, the output is often a binary variable. For example, consider a question – Is the mail spam or not spam? The answer is yes or no. Thus, logistic regression is used to predict categorical dependant variable.
9. Ridge regression is used to create a parsimonious model by introducing ridge estimator that introduces some bias in the model to reduce variance.
10. LASSO stands for “Least Absolute Shrinkage and Selection Operator”. This uses L_1 regularization to remove irrelevant variables.

Key Terms

- **Regression Analysis** – A mathematical technique of modelling the relationship between input and output variables.
- **Independent Variable** – The feature variables (x) in function $y = f(x)$ are called independent variables. These are also known as explanatory, exploratory, covariate or domain points.
- **Dependent Variable** – In the function, $y = f(x)$, y is called a dependent variable as it is dependent on x . Also known as labels or response variables.

- **Correlation** – Correlation indicates the strength of the relationship. Positive correlation among variables x and y is that they increase together in tandem. Negative correlation means as one variable increases, the other one decreases.
- **Scatter Plot** – A plot of explanatory variables and response variables.
- **Linearity** – The relationship between variables can be visualized as a straight line.
- **Non-linearity** – The relationship between variables is complex and cannot be fitted within a line.
- **Multicollinearity** – If the explanatory variables are highly correlated, then regression is vulnerable to bias. This condition is called Multicollinearity.
- **Singularity** – If the correlation among explanatory variables is perfect, it is called singularity.
- **Ordinary Least Square (OLS)** – Optimization technique where the distance between the data points and the line is optimized.
- **Best Fit** – The fitted line with the least sum of squared error is called best-fit.
- **Residual or Error** – The difference between the actual and predicted variables is called a residual.
- **Mean Absolute Error (MAE)** – The mean of residuals is called MAE.
- **Means Squared Error (MSE)** – The sum of square of residuals is called MSE.
- **r^2** – This is called coefficient of determination. This is square of the correlation between the predicted and response variables.
- **Multiple Regression** – The technique that finds the relationship between two or more independent variables and one dependent variable.
- **Polynomial Regression** – This is a technique of modelling using N^{th} order polynomial.
- **Odds** – The ratio of probability of an occurring event and a non-occurring event is called odds.
- **Logit** – Logit is log of odds. Otherwise, it is the log of outcome of logistic regression.
- **Logistic Function** – A logistic function is the one that takes a value and outputs either 0 or 1 by minimizing the error in the probability predicted by the model.
- **LASSO** – LASSO stands for ‘Least Absolute Shrinkage and Selection Operator’. This uses L_1 regularization.
- **Ridge Regression** – A regression technique that uses L_2 regularization.
- **Elastic Net** – A hybrid regression method of Ridge and LASSO regression.

Review Questions

1. What is the role of regression model in exploratory data analysis?
2. Distinguish between the terms: Classification, Regression and Estimation.
3. What is the difference between Classification and Regression models?
4. What is the principle of ordinary least square in linear regression? What are the pros and cons of regression models?
5. Consider the following dataset in Table 5.11 where the week and number of working hours per week spent by a research scholar in a library are tabulated. Based on the dataset, predict the number of hours that will be spent by the research scholar in the 7th and 9th week. Apply linear regression model.

Table 5.11: Sample Data

x_i (Week)	1	2	3	4	5
y_i (Hours Spent)	12	18	22	28	35

6. The height details of boys and girls are given in the Table 5.12.

Table 5.12: Sample Data

Height of Boys (x_i)	65	70	75	78
Height of Girls (y_i)	63	67	70	73

Fit a suitable line of best fit for the above data.

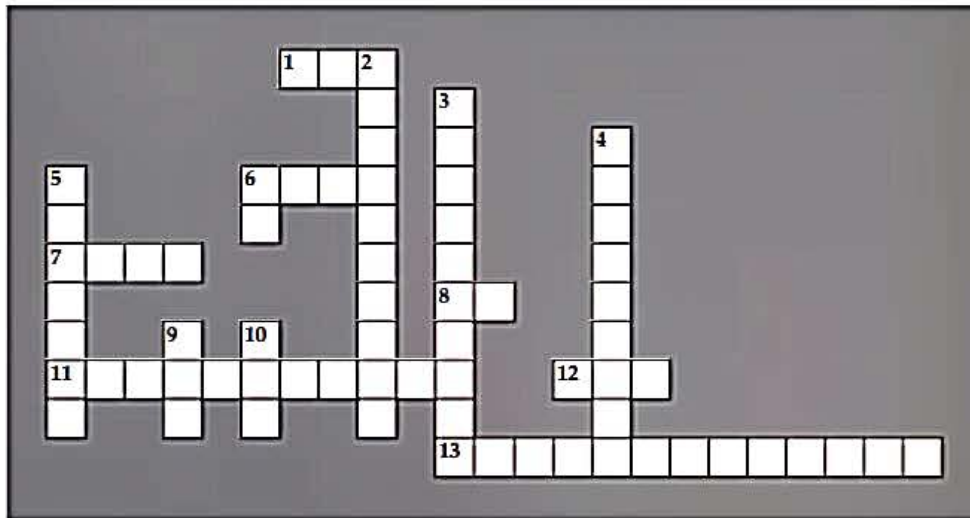
7. Using multiple regression, fit a line for the following dataset shown in Table 5.13. Here, z is the equity, x is the net sales and y is the asset. z is the dependent variable and x, y are independent variables. All the data is in million dollars.

Table 5.13: Sample Data

z_i	x_i	y_i
4	12	8
6	18	12
7	22	16
8	28	36
11	35	42

8. How does the polynomial regression model work?
9. How is logistic regression model different from linear regression?

Crossword



Across

1. Correlation measures the strength of the relationship. (Yes/No)
6. The linear relationship can be fitted using a _____.
7. The ratio of occurring and non-occurring events is called _____.
8. Ridge regression uses _____ regularization.
11. In the function $y = f(x)$, x is called _____ (Independent/Dependent) variable.
12. The average of absolute residuals is mean absolute error. (Yes/No)
13. The square of the correlation between the predicted and response variables is called coefficient of _____.

Down

2. Regression analysis is an example of _____ (supervised/unsupervised) learning.
3. In multi-collinearity, all variables are _____.
4. Polynomial regression is used to model _____ (Linear/Non-linear) relationships among variables.
5. The line that is fitted with the least sum of squares is called a line of _____.
6. Lasso uses _____ regularization.
9. Residual means the difference of predicted and actual variables. (Yes/No)
10. Scatter plot can be used to assess the relationship among variables. (Yes/No)

Word Search

Find and mark the words listed below.

D	K	R	C	G	N	E	C	G	O	B	R	C	H	A	I	B	I	X	J	G	V	D	C	F
R	G	V	K	R	G	X	C	D	X	I	S	Q	F	C	X	A	H	P	S	G	G	E	W	W
B	E	W	Y	E	S	K	E	F	I	W	O	T	V	O	U	Q	D	G	X	J	Y	T	E	D
Z	F	Q	V	Y	Z	S	S	O	C	D	D	V	D	R	B	W	V	L	S	V	V	E	C	I
G	L	E	P	D	L	V	U	J	S	T	L	I	R	R	Z	P	A	L	H	L	C	R	G	L
S	O	R	D	U	P	W	S	P	Q	E	G	P	V	E	K	N	Q	L	G	O	O	M	Y	Z
G	X	E	Q	G	N	E	B	F	E	Y	L	Y	N	L	O	F	Q	V	Z	J	J	I	C	H
X	X	O	E	Z	Y	P	I	E	A	R	R	Y	T	A	B	L	X	X	E	E	Y	N	L	K
P	O	A	N	Z	Y	W	K	S	V	J	V	X	C	T	G	V	L	Y	X	F	E	A	D	R
X	G	F	A	B	S	I	K	U	R	M	W	I	H	E	H	B	I	P	H	N	S	T	X	G
Z	Q	K	D	J	C	T	H	A	U	M	R	F	S	D	Y	D	N	A	P	O	E	I	N	O
H	F	Q	L	R	E	G	S	N	V	R	L	F	U	E	C	B	E	H	L	Z	Q	O	R	O
H	N	R	D	K	B	X	Z	K	T	A	B	S	F	N	D	D	S	H	K	L	U	N	I	D
L	W	S	X	B	A	I	A	R	K	S	D	T	C	S	T	M	J	T	S	L	O	G	R	F
B	D	Y	Q	Z	J	W	H	G	E	D	E	O	G	N	O	N	L	I	N	E	A	R	P	I
T	D	D	S	V	Q	T	Z	Y	O	C	L	Y	L	H	V	N	B	C	P	X	C	G	F	T
Y	K	F	U	Z	W	R	T	H	I	N	D	E	P	E	N	D	E	N	T	L	Z	N	X	K

- Independent

Good Fit

L1
- Supervised

Yes

L2
- Yes

Yes
- Yes

Determination
- Line

Nonlinear
- Correlated

Odds